

# Ratios, Proportion & Percentages



## Simplifying ratios

- 1 Simplify the ratio 18 : 24.
  - 2 Simplify the ratio 45 : 60 : 75.
  - 3 Simplify the ratio 36 : 8.
  - 4 Express "12 boys to 18 girls" as a ratio in simplest form.
- 

## Sharing a quantity in a given ratio

- 5 Share 150 in the ratio 2 : 3.
  - 6 Share 220 among three people in the ratio 3 : 4 : 4.
  - 7 A prize of 900 riyals is shared between Sara and Layla in the ratio 5 : 4. Find each share.
  - 8 Divide 84 kg of flour in the ratio 1 : 2 : 4.
- 

## Equivalent ratios and proportions

- 9 If  $3 : 5 = x : 20$ , find  $x$ .
  - 10 If  $a : 7 = 12 : 21$ , find  $a$ .
  - 11 A map has scale 1 : 50,000. A distance of 3 cm on the map represents how many meters in reality?
  - 12 Solve for  $y$ :  $\frac{4}{9} = \frac{y}{63}$ .
- 

## Percent of a number

- 13 Find 35% of 240.
- 14 Find 12% of 150.
- 15 What is 8% of 650?

- 16 Find 150% of 40.
- 

### Converting between fractions, decimals and percentages

- 17 Write  $\frac{3}{8}$  as a percent.
- 18 Write 0.45 as a fraction in lowest terms.
- 19 Write 62.5% as a fraction in lowest terms.
- 20 Write  $\frac{7}{40}$  as a decimal.
- 

### Percentage increase and decrease

- 21 Increase 80 by 25%.
- 22 Decrease 150 by 20%.
- 23 A salary of 4,000 riyals increases by 6%. Find the new salary.
- 24 A population of 2,500 decreases by 12%. Find the new population.
- 

### Finding percentage change and original values

- 25 A price rises from 80 to 92. Find the percentage increase.
- 26 A quantity falls from 250 to 200. Find the percentage decrease.
- 27 After a 20% discount, a jacket costs 96 riyals. Find the original price.
- 28 After a 15% increase, a value becomes 138. Find the original value.
- 

### Successive percentage changes

- 29 A price of 100 rises 10%, then falls 10%. Find the final price.
- 30 A stock worth 200 riyals rises 20% one year, then rises a further 25% the next year. Find the final value.
-

- 31 A quantity decreases by 10%, then decreases by a further 10%. Find the overall percentage decrease from the original amount.
- 32 A price of 400 riyals increases by 5%, then decreases by 5%. Find the final price, and state whether it equals the starting price.
- 

### Direct proportion

- 33  $y$  is directly proportional to  $x$ , and  $y = 15$  when  $x = 3$ . Find  $y$  when  $x = 8$ .
- 34 4 kg of apples cost 28 riyals. Find the cost of 7 kg at the same rate.
- 35 A car uses 8 liters of fuel to travel 96 km. How far can it travel on 20 liters at the same rate?
- 36 6 identical pens cost 27 riyals. Find the cost of 10 pens.
- 

### Inverse proportion

- 37 4 workers can complete a job in 15 days. How many days would 6 workers take, working at the same rate?
- 38 A car covers a fixed distance in 5 hours at 60 km/h. How long would the same trip take at 75 km/h?
- 39 8 taps fill a pool in 12 hours. How long would 3 taps take, running at the same rate?
- 

### Best value and unit price comparisons

- 40 A 3 kg bag of rice costs 27 riyals, and a 5 kg bag costs 42.50 riyals. Which is the better value per kilogram?
- 41 Store A sells 6 pens for 15 riyals; Store B sells 8 pens for 19.20 riyals. Which store offers the lower price per pen?
- 42 A 2-liter bottle of juice costs 14 riyals. Find the price per 100 mL.
- 43 Which is the better deal: 12 eggs for 9 riyals, or 18 eggs for 12.60 riyals?
-

## Word problems

- 44 A recipe requires flour and sugar in the ratio 5 : 2. If 350 g of flour is used, how much sugar is needed?
- 45 A shop raises the price of a shirt from 60 to 75 riyals, then during a sale reduces the new price by 20%. Find the sale price.
- 46 In a class of 40 students, 60% are girls. Then 4 more girls join the class. What percent of the class is now girls, to the nearest percent?
- 47 A dealership reduces the price of a car by 15% to 25,500 riyals. Find the original price.
- 48 Ahmed invests 8,000 riyals in an account that grows by 5% each year. Find its value after 2 years.
- 49 A map has scale 1 : 25,000. Two towns are 8.4 cm apart on the map. Find the actual distance between them in kilometers.
- 

## MCQ — Simplifying & writing ratios

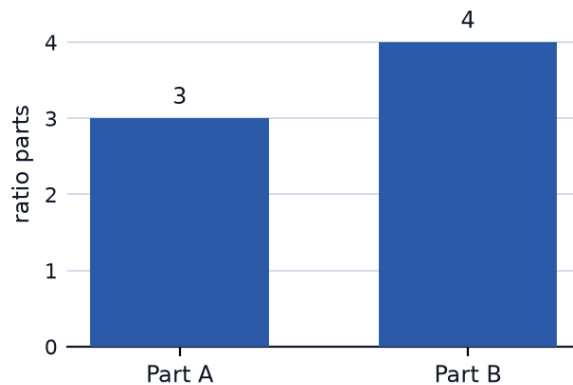
- 50 Simplify the ratio 42 : 56.
- a. 3 : 4
  - b. 6 : 8
  - c. 7 : 8
  - d. 3 : 8
- 51 Simplify 18 : 45.
- a. 3 : 5
  - b. 2 : 9
  - c. 2 : 5
  - d. 6 : 15

- 52 A recipe uses 250 g flour and 2 eggs. Write the ratio of flour (in grams) to eggs in simplest form.
- a. 1 : 125
  - b. 250 : 2
  - c. 125 : 1
  - d. 2 : 250
- 53 Express 0.75 : 2 as a ratio of whole numbers in simplest form.
- a. 3 : 4
  - b. 3 : 8
  - c. 75 : 2
  - d. 75 : 200

---

### MCQ — Sharing in a given ratio

- 54 The bar chart shows the ratio in which a total of 84 is to be shared. Find the larger share.



- a. 36
- b. 21
- c. 48
- d. 63

- 55 Share 260 SAR in the ratio 2 : 3 : 5. Find the smallest share.
- a. 78 SAR
  - b. 52 SAR
  - c. 130 SAR
  - d. 65 SAR
- 56 Two business partners share a profit of 18,000 SAR in the ratio 5 : 7. Find the larger share.
- a. 10,500 SAR
  - b. 9,000 SAR
  - c. 7,500 SAR
  - d. 12,857 SAR
- 57 A line of length 36 cm is divided in the ratio 4 : 5. Find the length of the shorter piece.
- a. 16
  - b. 4
  - c. 9
  - d. 20
- 58 A class of 40 students has boys and girls in the ratio 3 : 5. How many boys are there?
- a. 25
  - b. 8
  - c. 24
  - d. 15

---

## MCQ — Equivalent ratios & scale

- 59 If  $a : b = 3 : 5$  and  $a = 18$ , find  $b$ .
- a. 30
  - b. 15
  - c. 10.8
  - d. 24
- 60 A map has scale  $1 : 50,000$ . A distance of 3 cm on the map represents how many km in reality?
- a. 15 km
  - b. 1.5 km
  - c. 150 km
  - d. 0.15 km
- 61 Which ratio is equivalent to  $4 : 9$ ?
- a.  $9 : 4$
  - b.  $20 : 45$
  - c.  $8 : 16$
  - d.  $16 : 37$
- 62 In a photo, a model that is 1.8 m tall appears 6 cm high. Using the same scale, how tall (in m) is a building that appears 40 cm high in the photo?
- a. 4.8
  - b. 24
  - c. 7.2
  - d. 12

---

## MCQ — Percent of a number

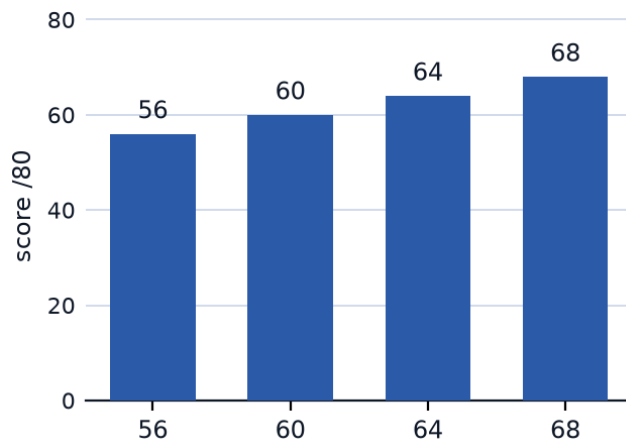
63 Find 35% of 260.

- a. 86
- b. 95
- c. 78
- d. 91

64 Find 12% of 450.

- a. 62
- b. 48
- c. 54
- d. 45

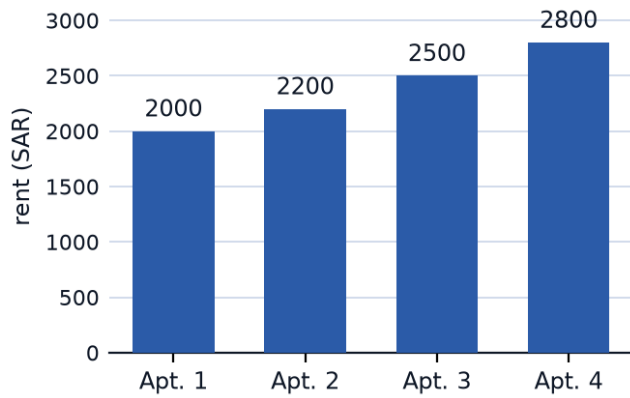
65 The bar chart shows four students' quiz scores out of 80. Which score is exactly 75%?



- a. 68
- b. 64
- c. 60
- d. 56



- 66 The bar chart shows monthly rent for four apartments. 8% of which apartment's rent equals 200 SAR?



- a. Apt. 2
  - b. Apt. 1
  - c. Apt. 3
  - d. Apt. 4
- 67 150% of a number is 300. Find the number.
- a. 250
  - b. 450
  - c. 200
  - d. 150

---

### MCQ — Fraction, decimal & percent conversion

- 68 Write  $\frac{7}{20}$  as a percent.
- a. 35%
  - b. 3.5%
  - c. 7%
  - d. 70%

- 69 Write 0.06 as a percent.
- a. 60%
  - b. 6%
  - c. 0.06%
  - d. 0.6%
- 70 Write 85% as a fraction in simplest form.
- a.  $\frac{17}{20}$
  - b.  $\frac{8}{5}$
  - c.  $\frac{17}{25}$
  - d.  $\frac{85}{100}$
- 71 Which of these is largest:  $\frac{3}{8}$ , 0.4, or 37%?
- a. 0.4
  - b. They are all equal
  - c.  $\frac{3}{8}$
  - d. 37%

---

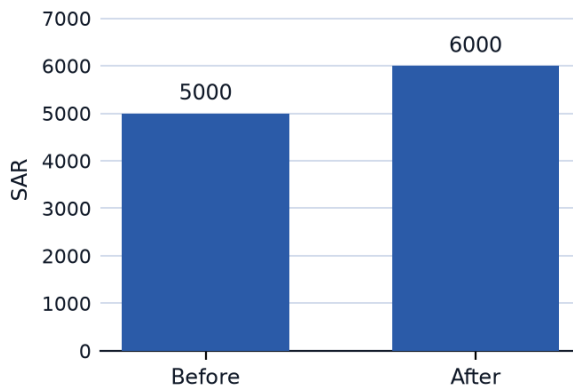
### MCQ — Percentage increase & decrease

- 72 Increase 80 by 25%.
- a. 105
  - b. 20
  - c. 100
  - d. 95

73 Decrease 150 by 12%.

- a. 138
- b. 128
- c. 132
- d. 18

74 The bar chart shows a salary before and after a raise. What percentage increase does it represent?



- a. 15%
- b. 500%
- c. 20%
- d. 25%

75 A shirt costs 120 SAR after a 20% discount. Find the original price.

- a. 150 SAR
- b. 144 SAR
- c. 100 SAR
- d. 140 SAR

**76** Find the percentage change from 40 to 34.

- a. 17.6% decrease
- b. 15% increase
- c. 6% decrease
- d. 15% decrease

---

### MCQ — Finding the original value

**77** After a 15% increase, a price is 92 SAR. Find the original price.

- a. 80 SAR
- b. 78.20 SAR
- c. 88 SAR
- d. 107 SAR

**78** After a 30% discount, an item costs 56 SAR. Find the original price.

- a. 72.80 SAR
- b. 80 SAR
- c. 86 SAR
- d. 74.67 SAR

**79** A population grew by 8% to reach 27,000. Find the population before the growth.

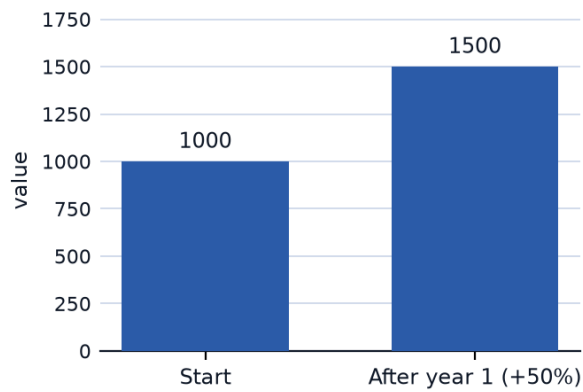
- a. 25,000
- b. 26,000
- c. 29,160
- d. 24,840

- 80 After including a 15% service charge, a restaurant bill is 138 SAR. Find the bill before the charge.
- a. 158.70 SAR
  - b. 120 SAR
  - c. 123 SAR
  - d. 117.30 SAR
- 

### MCQ — Successive percentage changes

- 81 A price of 200 rises 10%, then rises another 10%. Find the final price.
- a. 220
  - b. 240
  - c. 244
  - d. 242
- 82 A price of 500 falls 20%, then rises 20%. Find the final price.
- a. 460
  - b. 480
  - c. 500
  - d. 520

- 83 The bar chart shows a stock's value at the start and after rising 50% in year one. It then falls 50% in year two. Find its final value.



- a. 500
- b. 900
- c. 1,000
- d. 750

---

### MCQ — Direct proportion

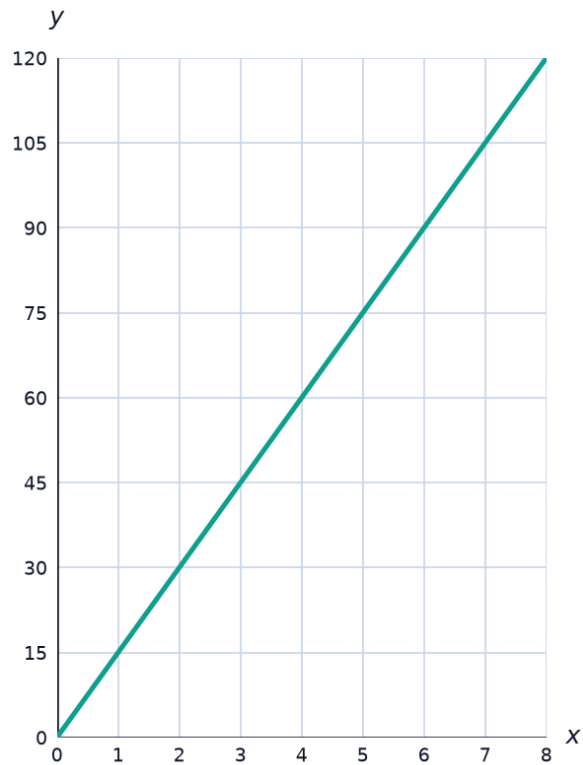
- 84 If 5 workers can build a wall in a task requiring 200 worker-hours total, and pay is directly proportional to hours worked, how much does one worker earn (in SAR) for 8 hours at a rate of 25 SAR/hour?

- a. 40
- b. 200
- c. 225
- d. 175

- 85 12 pens cost 30 SAR. At the same rate, find the cost of 20 pens.

- a. 45 SAR
- b. 60 SAR
- c. 40 SAR
- d. 50 SAR

- 86** The graph shows the direct proportion between the number of hours  $x$  worked and pay  $y$  in SAR,  $y = 15x$ . How much is earned for 6 hours?



- a. 105
- b. 90
- c. 75
- d. 21
- 87** A car uses 8 liters of fuel to travel 96 km. At the same rate, how far can it travel on 20 liters?
- a. 200
- b. 260
- c. 240
- d. 180

---

## MCQ — Inverse proportion

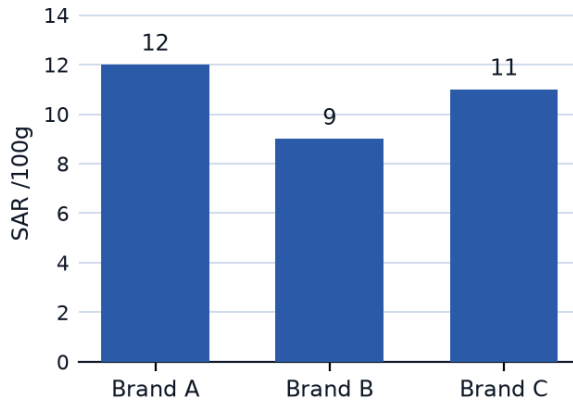
- 88 6 workers can finish a job in 10 days. How many days would 4 workers take (same rate each)?
- a. 15
  - b. 6.67
  - c. 24
  - d. 12
- 89 If  $y$  is inversely proportional to  $x$  and  $y = 8$  when  $x = 3$ , find  $y$  when  $x = 12$ .
- a. 0.5
  - b. 32
  - c. 4
  - d. 2
- 90 A tank is filled by 5 identical pumps in 12 hours. How long would 3 of the same pumps take?
- a. 15
  - b. 20
  - c. 7.2
  - d. 4

---

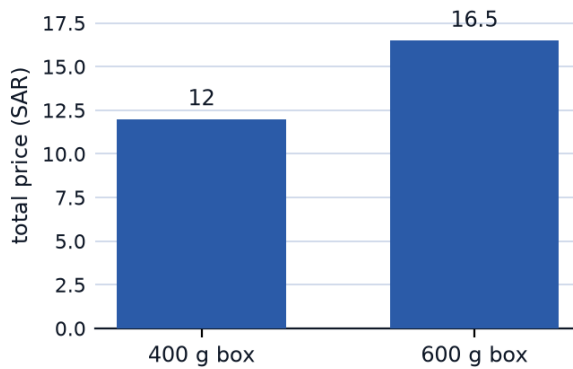
### MCQ — Best value & unit price



- 91 The bar chart shows the price per 100 g for three brands of coffee. Which brand offers the best value?



- a. Brand B
  - b. All are equal
  - c. Brand C
  - d. Brand A
- 92 The bar chart shows the total price of two cereal boxes of different sizes: 400 g and 600 g. Which has the lower price per 100 g?



- a. Cannot be determined
- b. They are equal
- c. The 600 g box
- d. The 400 g box

**93** Find the unit price (per kg) if 3.5 kg of rice costs 28 SAR.

- a. 7 SAR/kg
- b. 8 SAR/kg
- c. 9.80 SAR/kg
- d. 3.5 SAR/kg

**94** A 2 L bottle of juice costs 18 SAR. Find the price per 100 mL.

- a. 9 SAR
- b. 0.18 SAR
- c. 1.80 SAR
- d. 0.90 SAR

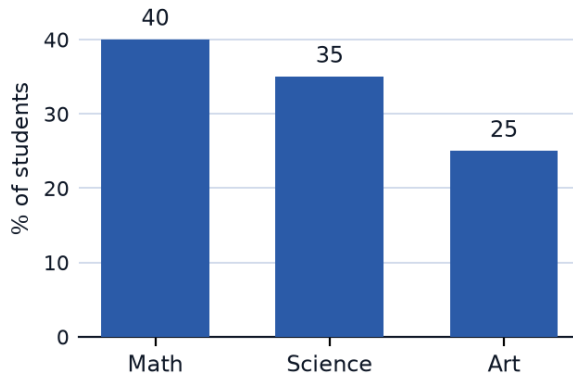
---

### MCQ — Word problems

**95** A car travels 240 km using 20 L of fuel. Find its fuel economy in km per liter.

- a. 14
- b. 10
- c. 20
- d. 12

- 96 The bar chart shows subject preference percentages from a survey of 250 students. How many students prefer math?



- a. 60
- b. 90
- c. 110
- d. 100
- 97 A recipe for 4 people needs 2.5 cups of rice. How many cups are needed for 10 people?
- a. 6.25 cups
- b. 6 cups
- c. 5 cups
- d. 25 cups
- 98 A company's revenue rose from 2.4 million to 3.0 million SAR. Find the percentage increase.
- a. 30%
- b. 25%
- c. 20%
- d. 60%

**99** A map scale is 1 : 200,000. Two cities are 8.5 cm apart on the map. Find the real distance in km.

a. 1.7

b. 34

c. 17

d. 8.5